FBG tilt angle prediction

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Aim and data

I predicted axial and transverse tilt separately from the two FBG wavelength changes. I used all 37 readings in each experimental sheet, including row 1. The theoretical sheet and extra chart cells were not used. No synthetic rows or imputations.

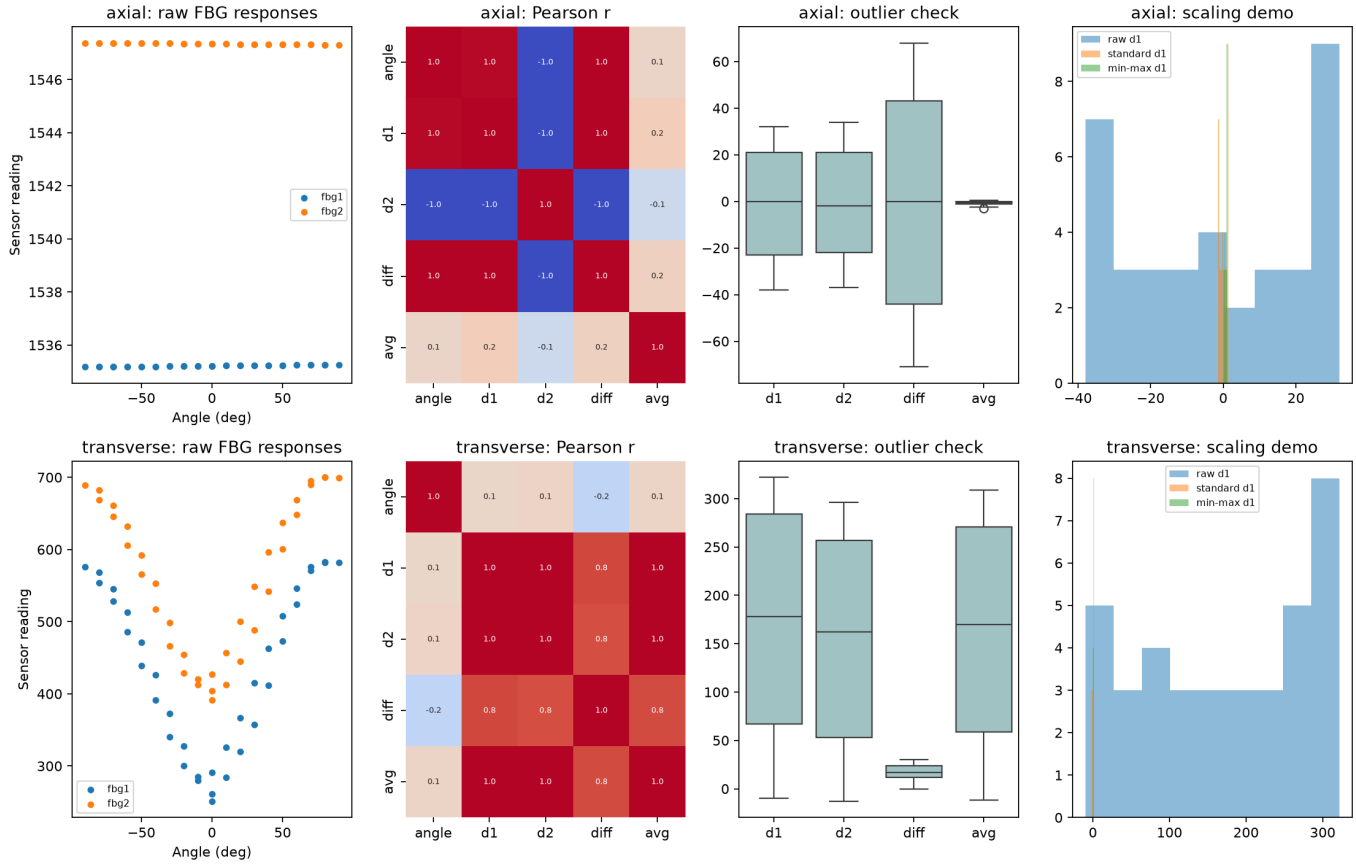
Axis	Rows	Angle range	Missing	Exact repeats	Repeated angles
Axial	37	-90 to +90 deg	0	2	17 values
Transverse	37	-90 to +90 deg	0	0	17 values

Method and one end-to-end model

F1 uses both wavelength changes, F2 their difference, F3 their average, and F4 difference plus average. I checked F2/F3 against the workbook columns. I used five outer GroupKFold splits by angle. Four grouped inner folds selected features and hyperparameters from training rows only. StandardScaler is inside each fitted pipeline. Metrics come from the joined outer-fold predictions.

Gaussian process example: D/E changes → F1/F2/F3/F4 → grouped split → inner RMSE chooses feature and RBF/Matérn kernel → fit scaler and GPR on outer training rows → predict held-out angles → repeat five times and calculate MAE, MSE, RMSE, R2. Seed 42 is used for stochastic trees.

Sensor responses, correlations, outliers and scaling



Axial difference tracks signed angle. In transverse tilt, both FBGs mostly rise with magnitude, so the sign is harder. Boxplots flag extremes but I kept all recorded measurements. Scaling shown here is only a demo; evaluation scalers fit training folds only.

Source: Kuo Li, *FBG biaxial tilt sensor using one optical fiber*, Mendeley Data, doi:10.17632/f4csv2xjk6.1, CC BY 4.0.

Feature check: grouped linear RMSE (degrees)

Axis	F1 pair	F2 diff	F3 average	F4 diff+avg
Axial	6.68	6.17	55.42	6.68
Transverse	46.54	53.89	53.29	46.54

F2 is strongest for axial linear prediction. Signed transverse prediction needs information from both FBGs; F1 and F4 contain the same two underlying measurements.

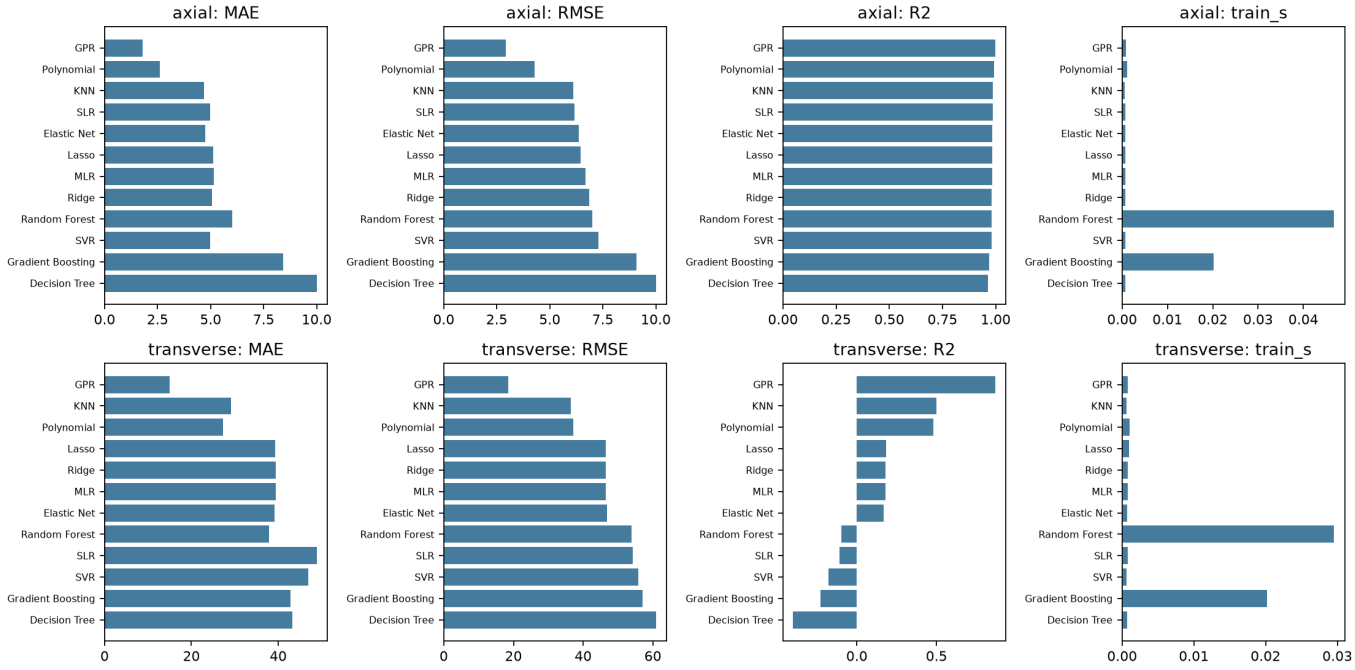
Nested-tuned performance: axial

Model	MAE	MSE	RMSE	R2	Fit ms	Feat.
GPR	1.80	8.6	2.93	0.997	0.82	F1
Polynomial	2.60	18.4	4.29	0.993	1.09	F2
KNN	4.68	37.2	6.10	0.986	0.62	F1
SLR	4.99	38.1	6.17	0.986	0.71	F2
Elastic Net	4.76	40.4	6.36	0.985	0.68	F1
Lasso	5.11	41.7	6.46	0.984	0.71	F2
MLR	5.15	44.7	6.68	0.983	0.71	F1
Ridge	5.06	46.9	6.85	0.982	0.73	F2
Random Forest	6.02	49.2	7.01	0.981	46.79	F1
SVR	4.98	53.3	7.30	0.980	0.71	F1
Gradient Boosting	8.41	82.5	9.08	0.969	20.15	F1
Decision Tree	10.00	100.0	10.00	0.962	0.71	F2

Nested-tuned performance: signed transverse

Model	MAE	MSE	RMSE	R2	Fit ms	Feat.
GPR	14.95	343.8	18.54	0.870	0.82	F4
KNN	29.10	1324.7	36.40	0.499	0.61	F4
Polynomial	27.33	1377.7	37.12	0.479	1.05	F1
Lasso	39.24	2158.1	46.46	0.184	0.99	F4
Ridge	39.41	2165.2	46.53	0.181	0.78	F4
MLR	39.48	2166.1	46.54	0.181	0.79	F4
Elastic Net	39.15	2194.4	46.84	0.170	0.68	F4
Random Forest	37.82	2903.9	53.89	-0.099	29.55	F4
SLR	48.85	2934.7	54.17	-0.110	0.76	F3
SVR	46.94	3117.9	55.84	-0.180	0.65	F4
Gradient Boosting	42.76	3246.3	56.98	-0.228	20.21	F4
Decision Tree	43.29	3701.9	60.84	-0.401	0.72	F2

Accuracy and fit-time comparison



All bars use outer-fold predictions after inner-fold selection. Fit time is the mean final training fit per fold, excluding search. On 37 rows it is only a rough speed indicator.

Hyperparameter checks: every requested setting

Fixed-setting scores below also choose features in the inner training folds. The final family scores on page 2 use inner-fold hyperparameter selection too; I did not choose a winner by looking at outer test scores.

Axial: RMSE and R2

Model	Configuration	RMSE	R2
SLR	default	6.17	0.986
MLR	default	6.68	0.983
Polynomial	degree=2	5.46	0.989
Polynomial	degree=3	4.29	0.993
Ridge	alpha=0.1	6.85	0.982
Lasso	alpha=0.1	6.46	0.984
Ridge	alpha=1	6.27	0.985
Lasso	alpha=1	6.56	0.984
Ridge	alpha=10	9.90	0.963
Lasso	alpha=10	12.46	0.941
Elastic Net	alpha=0.1, l1=0.5	6.36	0.985
Elastic Net	alpha=1, l1=0.5	12.62	0.940
SVR	kernel=linear, C=10, eps=0.1	7.30	0.980
SVR	kernel=rbf, C=10, eps=0.1	14.09	0.925
Decision Tree	depth=2	20.03	0.848
Decision Tree	depth=3	13.75	0.928
Decision Tree	depth=5	10.00	0.962
Random Forest	trees=50	7.18	0.980
Random Forest	trees=100	7.07	0.981
KNN	k=3	6.10	0.986
KNN	k=5	7.00	0.981
KNN	k=7	9.32	0.967
Gradient Boosting	lr=0.01, trees=100	18.88	0.865
Gradient Boosting	lr=0.1, trees=100	9.08	0.969
GPR	kernel=RBF	3.20	0.996
GPR	kernel=Matern	2.93	0.997

Transverse: RMSE and R2

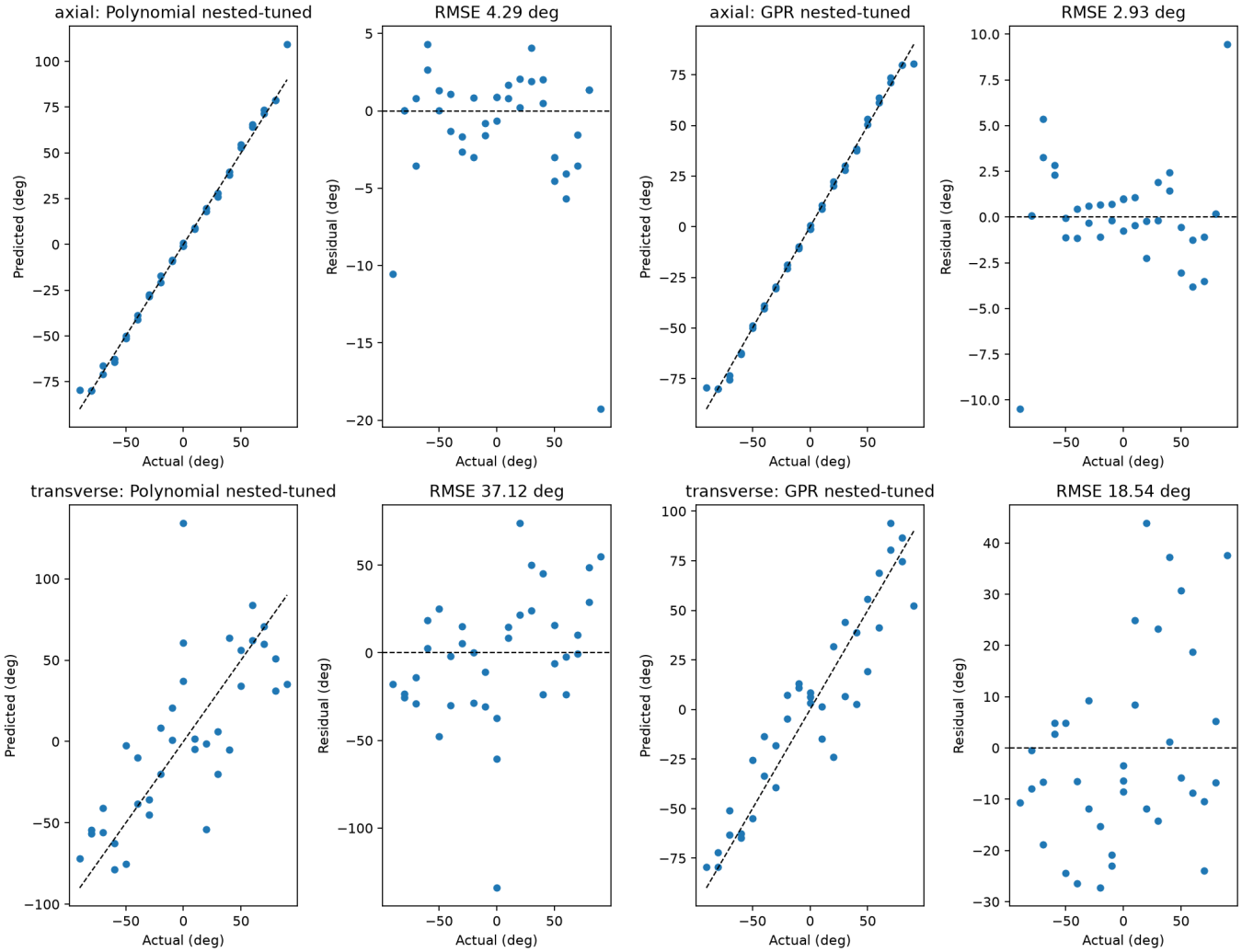
Model	Configuration	RMSE	R2
SLR	default	54.17	-0.110
MLR	default	46.54	0.181
Polynomial	degree=2	37.90	0.457
Polynomial	degree=3	38.88	0.428
Ridge	alpha=0.1	46.53	0.181
Lasso	alpha=0.1	46.46	0.184
Ridge	alpha=1	46.67	0.176
Lasso	alpha=1	46.65	0.177
Ridge	alpha=10	49.63	0.068
Lasso	alpha=10	52.58	-0.046
Elastic Net	alpha=0.1, l1=0.5	46.84	0.170
Elastic Net	alpha=1, l1=0.5	50.56	0.033
SVR	kernel=linear, C=10, eps=0.1	63.94	-0.547
SVR	kernel=rbf, C=10, eps=0.1	54.86	-0.138
Decision Tree	depth=2	56.05	-0.189
Decision Tree	depth=3	62.41	-0.474
Decision Tree	depth=5	64.17	-0.558
Random Forest	trees=50	53.97	-0.102
Random Forest	trees=100	45.71	0.210
KNN	k=3	31.77	0.618
KNN	k=5	44.12	0.263
KNN	k=7	49.51	0.073
Gradient Boosting	lr=0.01, trees=100	56.93	-0.226
Gradient Boosting	lr=0.1, trees=100	58.78	-0.307
GPR	kernel=RBF	20.96	0.834
GPR	kernel=Matern	17.00	0.891

Where the best ML model missed

Axis	Actual	Predicted	Abs. error	Fold
Transverse	20	-23.8	43.8	4
Transverse	90	52.5	37.5	4
Transverse	40	2.7	37.3	2
Transverse	50	19.3	30.7	5
Axial	-90	-79.5	10.5	5
Axial	90	80.6	9.4	4
Axial	-70	-75.3	5.3	2
Axial	60	63.8	3.8	4

These are the four largest held-out absolute errors for GPR on each axis. They show which individual angles are hard rather than hiding them in one average.

Actual versus predicted angles and residuals



Best statistical model: Polynomial on axial (4.29 deg RMSE), Polynomial on transverse (37.12 deg). Best conventional ML model: GPR on axial (2.93 deg), GPR on transverse (18.54 deg). These are held-out predictions.

Advanced task: signed versus absolute transverse tilt

Target	Input/model	MAE	RMSE	R2
Signed angle	Average change / linear	45.48	53.29	-0.074
Absolute angle	Average change / linear	5.60	6.75	0.937

The average change tells me the amount of transverse tilt much better than its direction. Positive and negative angles often produce similar readings. GPR finds some direction clues from both channels, but I would check those on more sensor runs before trusting the sign.

What I take from it

Degree 3 polynomial improves axial accuracy over a straight line, but added complexity is not always useful: deeper trees and more forest trees do not consistently win. Feature choice matters: axial F2 is useful alone, while signed transverse F3 is weak alone. Axial errors are smaller because its differential response tracks sign more directly.

For best accuracy on these measurements I would use Matérn GPR. For a simple real-time axial monitor, I would also consider linear regression on the differential signal because it is easy to calibrate and fast. For transverse monitoring I would report magnitude unless direction is separately validated. This is still only 37 readings per axis.

github.com/mdminhaj-2106/ML_Lab9 has the code, notebook and row-wise results.